Min Wang

Curriculum Vitae

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Education and Research

2015–2020 **Ph.D. in Mathematics**, *Department of Mathematics, University of Leicester*, UK.

2012–2013 **M.Sc. in Financial Mathematics and Computation**, *Department of Mathematics, University of Leicester*, UK.

2004–2008 **B.Eng.(Hons) in Software Engineering**, *School of Computer Science, Sichuan Normal University, Cheng Du*, China.

Research Interests

- My main interest remains in the areas of the intersection of probability and combinatorics, analytical combinatoric. My research has focused on developing combinatorial structure and path calculation methods and then apply them to model individual share price path and calculate option prices. The share price can be modelled as a path with a given share price changes and the expiry date.
- My future research is to build on the foundations of my PhD and to investigate algorithms related to combinatorial problem and graph construction, particularly interested in projects containing more programming tasks with MATLAB or C++. I also look forward to join the projects related to machine learning algorithm and implementation.

Academic Publications

1. **Modelling of share price paths via matroids**.
   (Utev, Sergey and Wang, MIN, 2019) (International Conference Paper, Under Review)
   In this study, we have developed an algorithm to construct share price paths from general binary trees. We also have provided a new relationship between lattice integer points and tutte polynomials of transversal matroid. Use this new way of connection together with the algorithm, we are able to construct a share price path from a given connected lattice path matroid.

2. **Counting via Markov-type technique**.
   (Utev, Sergey and Wang, MIN, 2019) (worked paper)
   In this study, we have developed an unusual stochastic model using markov-type technique and use this model to solve the specified counting problem. In this model, we consider counting restricted paths in a discrete time model, motivated by markov chain knowledge, we develop a new transform probability matrix, the matrix with different dimensions according to different restrictions can solve different counting problem. Considering the computer efficiency, we use sparse matrix to represent the transform matrix in MATLAB and solve the counting problem.

Some Working experience

9/2015–12/2019 **Teaching Assistant**, Department of Mathematics *and* School of Engineering , Univerisity of Leicester, UK.

Managing and Leading linear algebra feedback sessions for two classes, each containing up to 20 students. Assisting and providing learning support in numerous computer classes, ranging from the first year undergraduate to masters level, with topics covering scientific computing, differential equation, partial differential equation, excel, excel VBA, Matlab.

7/2016–8/2016 **Work Experience**, Financial Control and Asset Liability Department, ICBC(London) Plc. London, UK.

Studied annual financial reports and tax policy.

11/2015–6/2016 **Quiet and Library environment enforcement officer**, University of Leicester Library, Univerisity of Leicester, UK.

Patrolling the library areas to maintain a quiet, serious and calm study environment for all students. Providing library facility services for students or referring them to relevant library staff.

11/2011–1/2012 **Security Consultant Trainee**, Beijing Top-sec Network Security Technology Co., ChengDu, China.

Assisting the technologist to conduct risk assessments, perform physical network assessments and client services for Topsec security products.

5/2010–5/2011 **Principal**, Franchisee of Longman BACKPACK Engish, HeJin, China.

Organised and Founded Juvenile's English Training School. Recruiting English teachers and Organising training activities together with Beijing Longman education school. Setting the reasonable operational plan to run the school including managing budgets, arranging enrollments, evaluating teachers, actively contacting with local primary schools to provide teaching demonstrations.

Some Degree Courses Taken

2004-2008 Mathematics	Advanced Mathematics I-II(Calculus I-II), Linear Algebra, Probability Theory & Mathematical Statistics, Discrete Mathematics, College Physics(II) & Experiment
Computer Knowledge	Assembly Languages & Experiment, Principles of Computer Composition & Experiment, Operating System & Experiment, Principles of Network & Experiment, Computer Graphics, Network Programming, Information Security, Introduction to Software Engineering, Software Engineering, Software Testing Technology and Maintenance, Tools Software, Principles of Compiler
Operating Systems	Operating System & Experiment, Management and Application of Linux
Database, Data	Principles of Database, Management and Application of Linux, Oracle Technique of Database, Data Recovery
Algorithm	Data Structure, Data Structure Experiment, Algorithmic Design & Analysis
Programming	C Language Program Designing & Experiment, Object-Oriented Programming & Experiment(C++), Java Language Program Designing, VC Programming & Experiment, Webpage Design and Website Construction

2012-2013 Financial Mathematics I, Financial Mathematics II, Advanced C++ Programming, Scientific Computing(including computer assignment with MATLAB), Financial Risk, Computational Partial Differential Equations(PDE) with Applications(including computer assignment with MATLAB)

2015-2016 Financial Mathematics, Computational Partial Differential Equations(PDE) with Applications, Stochastic Process, Analytical Combinatoric, Abstract Algebra(online)

Some External Courses Taken

1–3/2020 C/C++ - Object-Oriented Programming Using C++ Part 2, City, University of London, UK

4–6/2019 Academic English for Postgraduate Research Students & Staff (CLS&CSE), University of Leicester, UK

2–5/2019 Presentation Skills for Research Students, University of Leicester, UK

11/2017 Training seminar "Preparing to Teach in Higher Education", University of Leicester, UK

9/2017 Teaching training seminar, Department of Mathematics, University of Leicester, UK

7–12/2014 Academic English Course, University of Leicester, UK

2–6/2014 CFA I, London School of Business, UK

Academic Teaching

12/2019 **System Dynamics and Control MATLAB Sessions(Year 2)**, School of Engineering, University of Leicester, UK

1–3/2019 **Computational Partial Differential Equations with Applications computer class, MATLAB(Master)**, Department of Mathematics, University of Leicester, UK

1–3/2019 **Computational Partial Differential Equations with with Finite Elements computer class, MATLAB(Master)**, Department of Mathematics, University of Leicester, UK

9–12/2018 **1st year computer classes with MATLAB(Year 1)**, Department of Mathematics, University of Leicester, UK

9–12/2017, 2018, 2019 **Scientific Computing computer class MATLAB(Year 3 and Master)**, Department of Mathematics, University of Leicester, UK

10–12/2016 **Linear Algebra 1 Problem/Feedback sessions (Year 1)**, Department of Mathematics, University of Leicester, UK

10–12/2016 **1st year computer classes with Excel VBA (Year 1)**, Department of Mathematics, University of Leicester, UK

1–3/2016 **Differential Equations and Dynamics computer class, MATLAB (Year 2)**, Department of Mathematics, University of Leicester, UK

10–12/2015 **1st year computer classes with Excel (Year 1)**, Department of Mathematics, University of Leicester, UK

2015-2019 **Marking all assignments, project submitted and providing constructive feedback in the teaching roles**, Department of Mathematics, University of Leicester, UK

Selected Projects

2015–2019 **Path Calculations and Option Pricing**, Prof. Sergey Utev, University of Leicester, UK

9/2013 **Restoring Utility Function Or Risk Preference**, Dr. Bogdan Grechuk, University of Leicester, UK

7/2008 **Data Warehouse Design and Management**, Prof. XiaoNing Li, Sichuan Normal University, China

Technical skills

Excel VBA, MATLAB, Latex, C++, Git, GitHub, ATOM

Some Achievements from Projects and Courses

- Awarded B.Eng. Bachelor degree, M.Sc. Mathematics Master degree, Ph.D. in Mathematics Doctoral degree
- Received Second-top academic scholarship each semester in every academic year, Department of Computer Science, at Sichuan Normal University, China
- Financial Mathematics II (grade: A), Scientific Computing assignments average over 65, PDE computer assignments average over 75, Department of Mathematics, at University of Leicester
- Planned an oral presentation and Give a presentation to audiences in the ELTU PGR Spring Lecture Series, University of Leicester

Some Skills from Projects, Teaching and Courses

- Solid knowledge of software engineering and strong ability to ready to master the programming languages
- Understanding combinatorial questions, algorithms and techniques
- Modelling discrete time share price paths using combinatorial structures including tree structures, matroid and polyomino
- Stochastic Modelling discrete time share price paths using Markov Chain-type techniques and applying the model to solve combinatorial problems
- Representing discrete time share price paths using generating functions and symbolic methods
- Numerical Analysis Programming using MATLAB including solving systems of linear and nonlinear equations, interpolation, least square, iterative solver and simple error analysis
- Solving systems of PDEs analytically and computationally using finite difference methods
- Analysing data using Excel and Excel VBA
- Constructing algorithms in the computer programming languages

- Programming using C++ and mastering the fundamental technical knowledge of C++ including object-oriented programming, pointers, references, some knowledge of error handling and template
- Creating and compiling C++ programs on the command line using g++ and editor ATOM
- Version control on projects using Git and GitHub on command line

Academic Administration

2015, 2019 Assisted in examining and invigilating examinations, Department of Mathematics, University of Leicester, UK

Conference and Seminars

4/2020 Modelling of share price paths via matroids, ICACSM 2020: 14. International Conference on Analytic Combinatorics and Symbolic Methods, London, UK

3/2019 Path Calculation and Option Pricing, PhD seminar, University of Leicester, UK (Oral Presentation)

3/2019 Path Calculation and Option Pricing, Doctoral College Poster Fairs, University of Leicester, UK (Poster Presentation)

8/2015 The 38th Research Students' Conference (RSC), University of Leeds, UK (Session Chair)

Languages

- Chinese, English

Volunteer

2017-present Mandarin Chinese Teaching: Children 5–8 year old.

Interests & Hobbies

- Sports (playing squash, cycling, climbing), Traveling, Cooking

Referees

- Prof Sergy Utev, Professor of Mathematics
  Department of Mathematics, University of Leicester
  su35@leicester.ac.uk
  +44(0)116 223 1310
- Bo Wang, Associate Professor in Statistics
  Department of Mathematics, University of Leicester
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- XiaoNing Li, Associate professor of Computer Science
  College of Computer Science, Sichuan Normal University
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